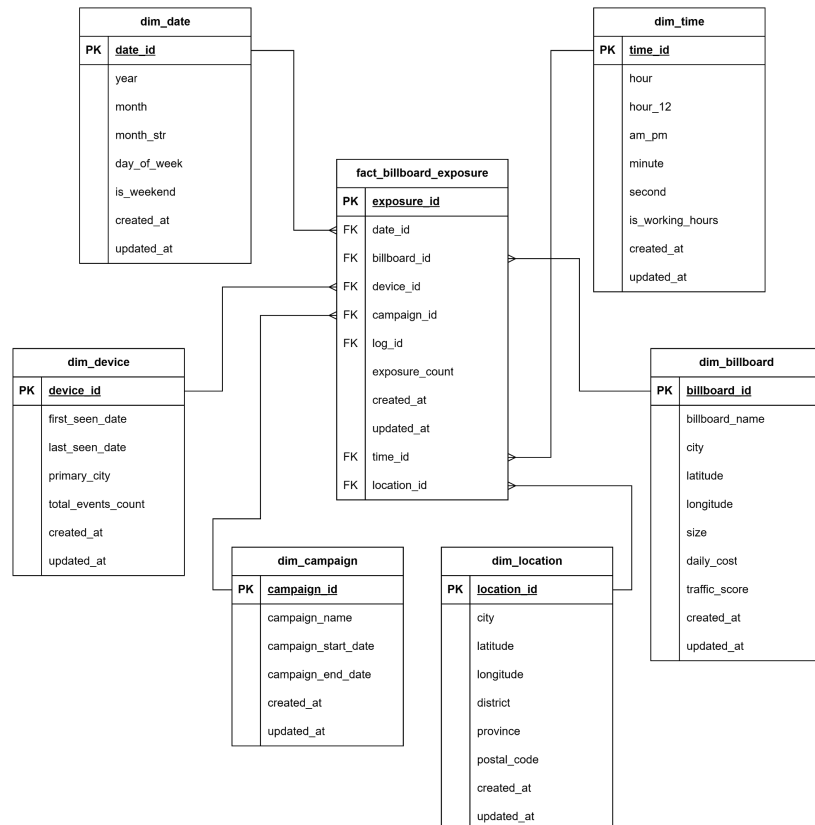


A. ERD

Grain yang dipakai: satu baris = satu exposure event => satu device melihat satu campaign dalam satu billboard pada satu waktu. Beberapa alasan kenapa menggunakan model ini:

- dapat dilakukan aggregate ke level apapun, seperti hourly, daily, monthly, per billboard dkk
- Bisa drill-down detail
- kebutuhan sekarang untuk billboard reach (daily/monthly) dapat dipenuhi

Berikut merupakan ERD sederhana menggunakan star schema.



B. Table Structures

Berikut merupakan tabel struktur nya:

1. dim_billboard

```

billboard_id STRING PRIMARY KEY
billboard_name STRING NOT NULL
city STRING NOT NULL
latitude FLOAT NOT NULL
longitude FLOAT NOT NULL
size STRING NOT NULL (CHECK IN ('Small', 'Medium', 'Large'))
daily_cost INTEGER NOT NULL (CHECK > 0)
traffic_score INTEGER NOT NULL (CHECK BETWEEN 1 AND 100)
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

INDEX idx_city (city)
  
```

2. dim_campaign

```
campaign_id STRING PRIMARY KEY
campaign_name STRING NOT NULL
campaign_start_date DATE NOT NULL
campaign_end_date DATE NOT NULL
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

INDEX idx_date_range (campaign_start_date,
campaign_end_date)
CHECK (campaign_end_date >= campaign_start_date)
```

3. dim_device

```
device_id STRING PRIMARY KEY
first_seen_date DATE NOT NULL
last_seen_date DATE NOT NULL
primary_city STRING
total_events_count INTEGER NOT NULL DEFAULT 0
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

INDEX idx_primary_city (primary_city)
INDEX idx_last_seen (last_seen_date)
```

4. dim_location

```
location_id STRING PRIMARY KEY
city STRING NOT NULL
latitude FLOAT NOT NULL
longitude FLOAT NOT NULL
sub_district STRING NULL
district STRING NULL
province STRING NULL
postal_code STRING NULL
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

INDEX idx_city (city)
INDEX idx_coordinates (latitude, longitude)
```

5. dim_date

```
date_id INTEGER PRIMARY KEY
full_date DATE NOT NULL UNIQUE
day_of_week INTEGER NOT NULL (1-7, 1=Monday)
day_name STRING NOT NULL (Monday, Tuesday, ...)
is_weekend BOOLEAN NOT NULL
```

```

is_holiday BOOLEAN NOT NULL DEFAULT FALSE
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

INDEX idx_full_date (full_date)
INDEX idx_is_weekend (is_weekend)

```

6. dim_time

```

time_id INTEGER PRIMARY KEY
hour INTEGER NOT NULL (0-23)
hour_12 INTEGER NOT NULL (1-12)
am_pm STRING NOT NULL (AM/PM)
minute INTEGER NOT NULL (0-59)
second INTEGER NOT NULL (0-59)
is_working_hours BOOLEAN NOT NULL (08:00-17:00 = TRUE)
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

INDEX idx_hour (hour)
INDEX idx_working_hours (is_working_hours)

```

7. fact_billboard_exposure

```

exposure_id STRING PRIMARY KEY (auto-generated UUID)
date_id INTEGER NOT NULL (FK → dim_date)
time_id INTEGER NOT NULL (FK → dim_time)
billboard_id STRING NOT NULL (FK → dim_billboard)
device_id STRING NOT NULL (FK → dim_device)
campaign_id STRING NOT NULL (FK → dim_campaign)
location_id STRING NOT NULL (FK → dim_location)
log_id STRING NOT NULL (FK → campaign_logs source)
event_id STRING NOT NULL (FK → mobility_events source)
exposure_count INTEGER NOT NULL DEFAULT 1
created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

INDEX idx_date_billboard (date_id, billboard_id)
INDEX idx_date_campaign (date_id, campaign_id)
INDEX idx_device (device_id)
INDEX idx_campaign (campaign_id)
INDEX idx_log_event (log_id, event_id)

```

C. Partition & clustering strategy

No	Tables	Partition & Cluster	Alasan
1	fact_billboard_exposure	partition: created_at cluster: campaign_id, billboard_id, device_id	query hampir selalu filter by date dan ini adalah tabel terbesar dengan timeseries

			yang terus bertambah
2	dim_device	cluster: primary_city	clustering by primary_city mengelompokkan data geografis sehingga query city tertentu hanya scan subset kecil
3	dim_billboard	cluster: city, location_id	clustering by city mengelompokkan billboard per kota untuk geo-analytics
4	dim_campaign	cluster: campaign_start_date, campaign_end_date	query sepertinya akan sering filter by date range untuk campaign reporting
5	dim_location	cluster: city, latitude, longitude	clustering by city mengelompokkan billboard per kota untuk geo-analytics